

Caroline Mazini Rodrigues, PhD

Researcher in Explainable AI and Deep Learning

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Research Experience

Postdoctoral Researcher

IRISA – CNRS

02/2025 – Present

France

- SHARP – PEPR-IA project with Nicolas Keriven and Thomas Maugey
- Interpretability of deep image compression models
- Analysis of latent representations in learned compression systems
- Frugal AI methods for reducing model complexity and computational cost

Research and Teaching Associate (ATER)

Université Gustave Eiffel

09/2023 – 09/2024

France

- CNN interpretability using hierarchical image segmentation
- Facilitating human analysis of image regions influencing biased model decisions

PhD Researcher in Explainable AI

LIGM (Université Gustave Eiffel) / LRE (EPITA)

09/2020 – 08/2023

France

- Research in explainable AI for computer vision models
- Improving visualization of gradient-based attribution methods using contrastive class information
- Global explanation of CNNs via concept decomposition

Data Scientist

Neuralmind

02/2020 – 08/2020

Brazil

- Development of deep learning models for computer vision and NLP tasks
- CNN-based object detection systems
- Transformer-based information extraction from unstructured text

Researcher in Complex Data Analysis

University of Campinas (Unicamp)

02/2018 – 08/2020

Brazil

- FAPESP-funded MSc fellowship at RECOD.ai
- Research on large-scale social media image analysis
- Representative image retrieval for forensic event analysis

Education

PhD in Computer Science

Université Gustave Eiffel

2020 – 2024

France

Supervisors: Prof. Laurent Najman (LIGM) and Dr. Nicolas Boutry (LRE)

MSc in Computer Science

University of Campinas (Unicamp)

2018 – 2020

Brazil

Supervisors: Prof. Zanoni Dias (LOCo) and Prof. Anderson Rocha (RECOD)

BSc in Computer Science

São Paulo State University (UNESP)

2013 – 2017

Brazil

Exchange semester at the University of Glasgow (2016)

Publications

Journal Articles.....

- C. M. Rodrigues, N. Boutry, L. Najman. *Explaining with Trees: Interpreting CNNs Using Hierarchies*. **Transactions on Machine Learning Research (TMLR)**, 2026.
- C. M. Rodrigues, N. Boutry, L. Najman. *Unsupervised Discovery of Interpretable Visual Concepts*. **Information Sciences**, 2024.
- C. M. Rodrigues, N. Boutry, L. Najman. *Transforming Gradient-Based Techniques into Interpretable Methods*. **Pattern Recognition Letters**, 2024.

- C. M. Rodrigues, A. Soriano-Vargas, L. Bahram, A. Rocha, Z. Dias. *Manifold Learning for Real-World Event Understanding*. **IEEE Transactions on Information Forensics and Security**, 2021.
- R. Padilha, C. M. Rodrigues, F. A. Andalo, G. Bertocco, Z. Dias, A. Rocha. *Forensic Event Analysis: From Seemingly Unrelated Data to Understanding*. **IEEE Security & Privacy**, 2020.

Conference Proceedings.....

- C. M. Rodrigues, N. Keriven, T. Maugey. *Efficient Training for Compact Compression Models via Sequential Distillation*. **European Signal Processing Conference (EUSIPCO)**, 2026.
- M. Doh, C. M. Rodrigues, N. Boutry, L. Najman, M. Mancas, H. Bersini. *Bridging Human Concepts and Computer Vision for Explainable Face Verification*. **BEWARE-23 Joint Workshop @ AIXIA**, 2024.
- C. M. Rodrigues, N. Boutry, L. Najman. *Gradients Intégrés Renforcés*. **Explain'AI – Conférence Francophone sur l'Extraction et la Gestion des Connaissances (EGC)**, 2023.
- C. M. Rodrigues, L. Pereira, A. Rocha, Z. Dias. *Image Semantic Representation for Event Understanding*. **IEEE International Workshop on Information Forensics and Security (WIFS)**, 2019.
- C. M. Rodrigues, M. A. Piteri, A. O. Artero, D. M. Eler, F. A. Silva, D. R. Pereira. *Facial Recognition in Digital Images using Local Binary Pattern Methods*. **XIII Workshop de Visão Computacional (WVC)**, 2017.

Invited Talks

- “Balancing fidelity and interpretability in XAI for global model understanding”.
Invited seminars at: LISN (03/2026), LIP6 (05/2026 – upcoming), GREYC (06/2026 – upcoming)
- “Explainable AI”. Invited seminar (part of AI for images course): ESIEE (10/2023)

Teaching

Course	Year	Type	Level	Institution	Hours
Images and Fundamentals	2023/24	TP	M1	UGE	12h
Algorithms and Programming	2023/24	TP/TD	L1	UGE	88h
Project Supervision	2023/24	TP	L1	UGE	14h
Databases	2023/24	TP	L2	UGE	24h
Introduction to C Programming	2023/24	TP	L2	UGE	24h
Python Big Data	2021–2023	TP	MSc	EPITA	48h
Signal Mathematics	2021–2023	TP/TD	MSc	EPITA	20h
Introduction to Neural Networks	2021/22	TD	MSc	EPITA	12h
Rational Languages Theory	2021/22	TP/TD	L1	EPITA	18h
Algorithms Complexity	2021/22	TD	L1	EPITA	14h

Scientific Activities

- Hi! PARIS Summer School – Institut Polytechnique de Paris (2025)
- Paris Open Source AI Summit (2025)
- Oxford Machine Learning Summer School – OxML (2022, 2024)
- EuADS Data Science for Explainable and Trustworthy AI (2023)
- Explain'AI Workshop at EGC (2023)
- KHIPU Latin American Meeting in AI (2019)

Awards

- Best Presentation – PhD Day MSTIC (2021)
- Academic Merit – São Paulo State University (2017)

Reviewer

ICLR (2024/25), NeurIPS (2026 – upcoming), EUSIPCO (2026), JVCIR – Elsevier (2023/25), NPL – Springer (2025), IJDSA – Springer (2026)

Languages

Portuguese (Native) English (Advanced) French (Advanced) Spanish (Basic) German (Basic – Learning)